

Customer Experience Intelligence

Analyzing Customer Satisfaction Patterns in E-Commerce Support Using Data Analytics

A Data Analytics Project

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Introduction & Problem Statement

E-commerce platforms generate large volumes of customer support data, but extracting meaningful insights from this data is challenging.

Project Objective

Analyze customer interaction data to identify factors influencing customer satisfaction and provide actionable insights for improving customer experience.



Dataset Overview

Dataset Name: **E-Commerce Customer Service Satisfaction Dataset**

Size:

- 85,907 customer interaction records
- 20 attributes

Important Features:

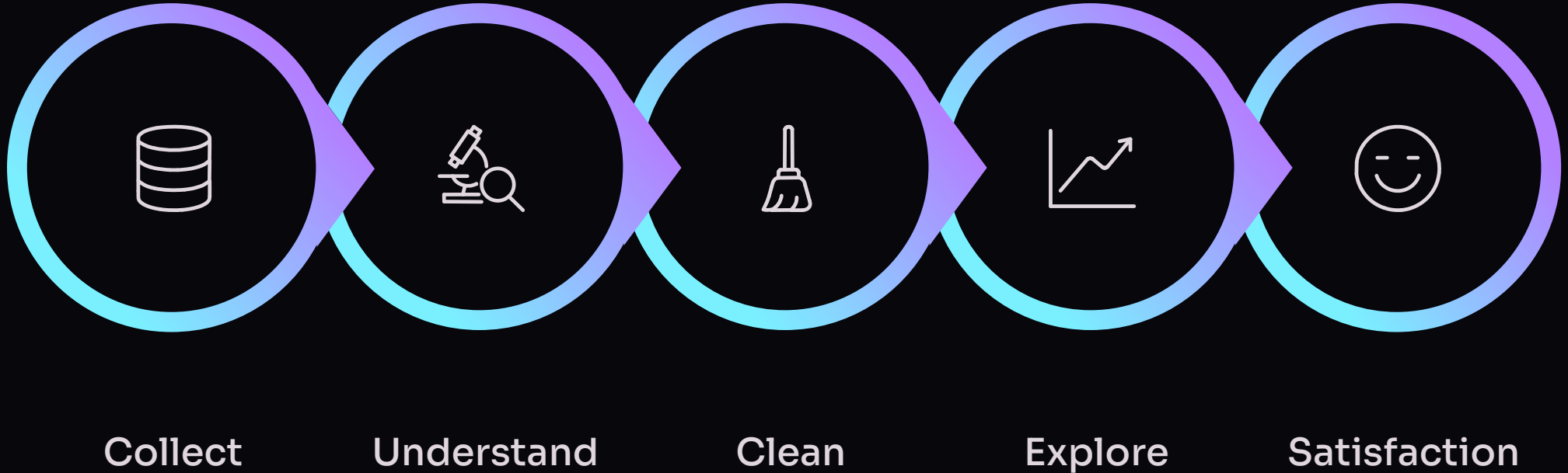
- Customer remarks
- Issue categories
- Response details
- Agent information
- Customer Satisfaction Score (CSAT)

Target Metric:

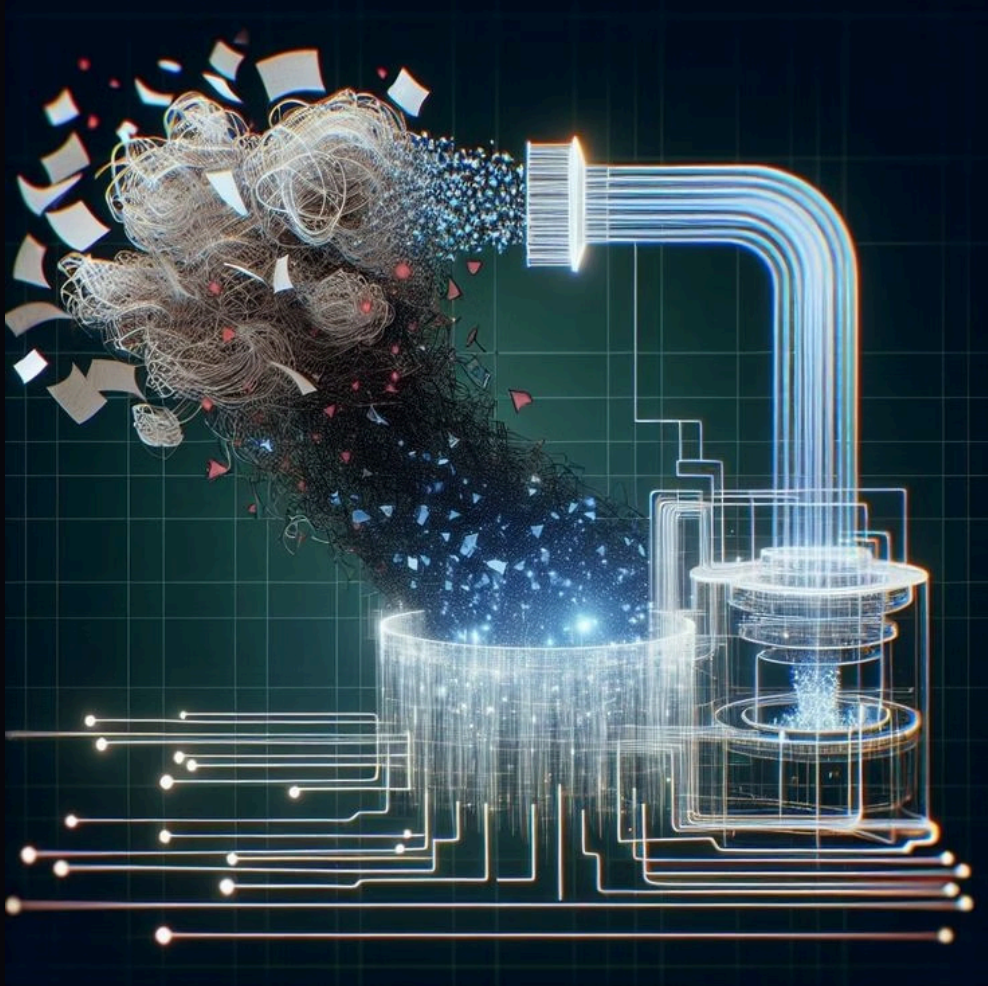
CSAT Score (1–5 rating)



Project Methodology



Each phase applied reproducible pipelines and validation checks to ensure data quality and reliable insights for stakeholders.

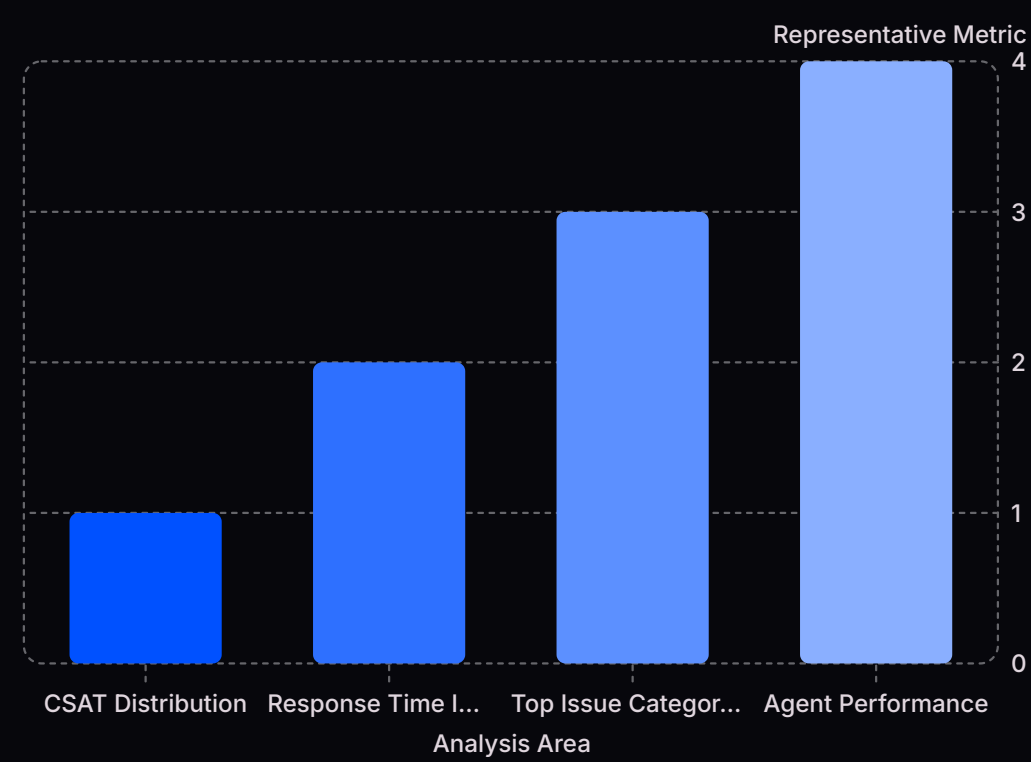


Data Cleaning & Preparation

- Key Steps

- Identified missing values and data quality issues
- Removed irrelevant attributes for analysis
- Standardized categorical and date formats
- Prepared customer interaction data for analysis
- Created meaningful analytical variables such as response time

Exploratory Data Analysis (EDA)



Customer Analysis

- CSAT score distribution
- Customer feedback patterns



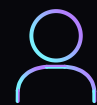
Service Analysis

Response time impact on satisfaction



Category Analysis

- Common customer issues
- High dissatisfaction areas



Performance Analysis

- Agent and service efficiency



Initial Analytical Findings

Response Time Matters

Faster first response and shorter resolution times are strongly correlated with higher CSAT.

Issue Categories Drive Dissatisfaction

Specific complaint types (e.g., delivery issues, returns, product defects) appear disproportionately in low-CSAT cases.

Feedback Reveals Pain Points

Text feedback highlights recurring themes: unclear policies, lack of proactive updates, and inconsistent agent communication.

Service Efficiency Influences Experience

High-performing agents combine quick responses with clear resolutions — both matter for loyalty.



Business Recommendations

- Improve response and resolution time
- Focus on frequently occurring customer issues
- Monitor service quality metrics
- Use customer feedback for continuous improvement

Future Enhancements



Sentiment Analysis

Apply NLP to extract emotions and granular pain points from open-text feedback.



CSAT Prediction

Build supervised models to predict satisfaction and enable proactive interventions.



Interactive Dashboards

Deploy Power BI dashboards for stakeholders with drilldowns by product, agent, and issue.



Real-Time Monitoring

Implement streaming analytics to flag at-risk interactions as they occur.





Conclusion

This project demonstrates how customer support data can be transformed into meaningful insights using analytics.

By analyzing customer interactions, satisfaction scores, and service patterns, organizations can identify improvement opportunities and make data-driven decisions to enhance customer experience.

Thank you. Questions?

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